

*Chapter No 3 – Hardware, Software & Networks***CE – 101 : C & P F***Computing and Programming
Fundamentals***Batch 2015**

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**CE - 101 : Computing and Programming
Fundamentals****■ Course Objectives:**

- Upon successful completion of this course, the student will be able to:
 - Fundamentals of Computer Engineering and Information Technology
 - Overview of History, Classification and Components of Computers
 - Number Systems and Logic Gates
 - Overview of Software, Operating Systems, Networks and Internet
 - Introduction about Programming
 - Basic Building Blocks, Loop, Decision Making

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C&PF Books

▪ Text Book:

1. **Introduction to Computers (7th Edition)**
By Peter Norton
2. **Turbo C Programming For The PC (Revised Edition)**
By Robert Lafore

▪ Reference Books:

1. **Computer, Communications and Information.**
By Sarah Hutchinson and Stacey Sawyer
2. **Let Us C**
By Yashavant Kanetkar

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Marks Distribution

▪ Mid Term	_____	20
▪ Lab Work	_____	20
▪ Quiz	_____	5
▪ Assignment + Class Performance	_____	5
▪ Semester Final Paper	_____	50
▪ Total Marks	_____	100
▪ Performance Bonus	_____	5 Marks

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Course Instructors

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Room No.BF-03
Section B

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Course Outline

✓ PART A

- History of Computer
- Classification of Computer
- Basic Components
- CPU, I/O, Peripheral Devices, Storage
- Von Neumann Architecture
- Number Systems
- Binary Numbers
- Boolean Logic
- Software
- Operating Systems
- Networks and Internet

✓ PART B

- Types of Programming Languages
- Algorithm
- Flow Chart
- Introduction to C Language Programming
- Basic Building Blocks
- Loops
- Decision Making Statements

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Chapter**3****Batch 2014**

Hardware, Software & Networks

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How Computers Represent Data

- Number systems
 - A manner of counting
 - Several different number systems exist
- Decimal Number system
 - Used by humans to count
 - Contains ten distinct digits
 - Digits combine to make larger numbers
- Binary number system
 - Used by computers to count
 - Two distinct digits, 0 and 1
 - 0 and 1 combine to make numbers

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How Computers Represent Data

■ Bits and Bytes

- Binary numbers are made of bits
- Bit represents a switch
- A byte is 8 bits
- Byte represents one character



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How Computers Represent Data

■ Text codes

- Converts letters into binary
- Standard codes necessary for data transfer
- ASCII
 - American English symbols
- Extended ASCII
 - Graphics and other symbols
- Unicode
 - All languages on the planet

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A Look Inside The Processor

- Architecture
 - Determines
 - Location of CPU parts
 - Bit size
 - Number of registers
 - Pipelines
 - Main difference between CPUs

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Microcomputer Processors

- Intel
 - Leading manufacturer of processors
 - Intel 4004 was worlds first microprocessor
 - IBM PC powered by Intel 8086
 - Current processors
 - Centrino
 - Itanium
 - Pentium IV
 - Xeon



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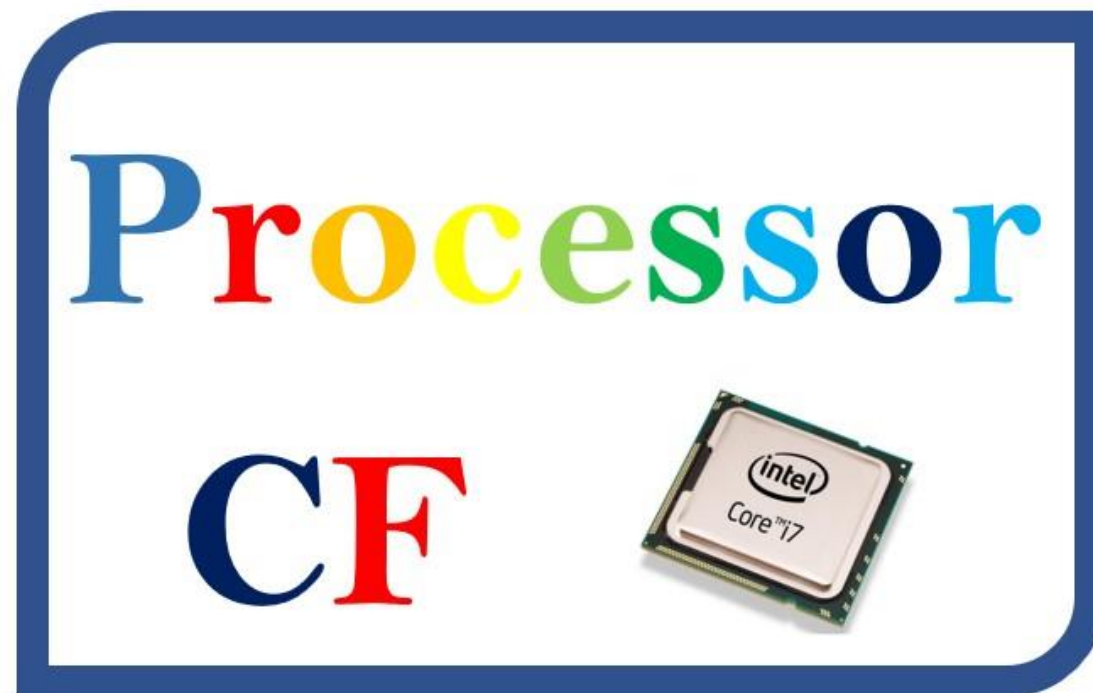
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Microcomputer Processors

View in AR



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Comparing Processors

- Speed of processor
- Size of cache
- Number of registers
- Bit size
- Speed of Front side bus

Table 5B.3 on page 210 lists the specifications for three powerful processors.

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Extending The Processors Power

- Standard computer ports
 - Keyboard and mouse ports
 - USB ports
 - Parallel
 - Network
 - Modem
 - Audio
 - Serial
 - Video

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Standard Computer Ports



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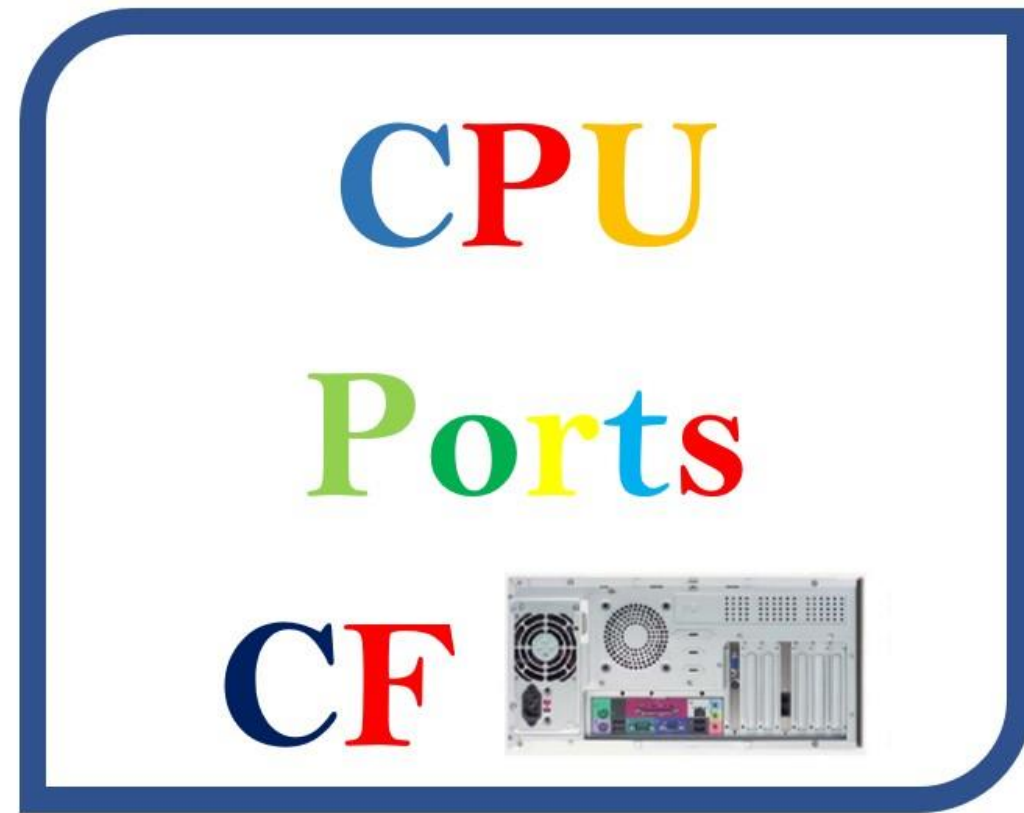
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Standard Computer Ports

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Extending The Processors Power

- Serial and parallel ports
 - Connect to printers or modems
 - Parallel ports move bits simultaneously
 - Made of 8 – 32 wires
 - Internal busses are parallel
 - Serial ports move one bit
 - Lower data flow than parallel
 - Requires control wires
 - UART converts from serial to parallel

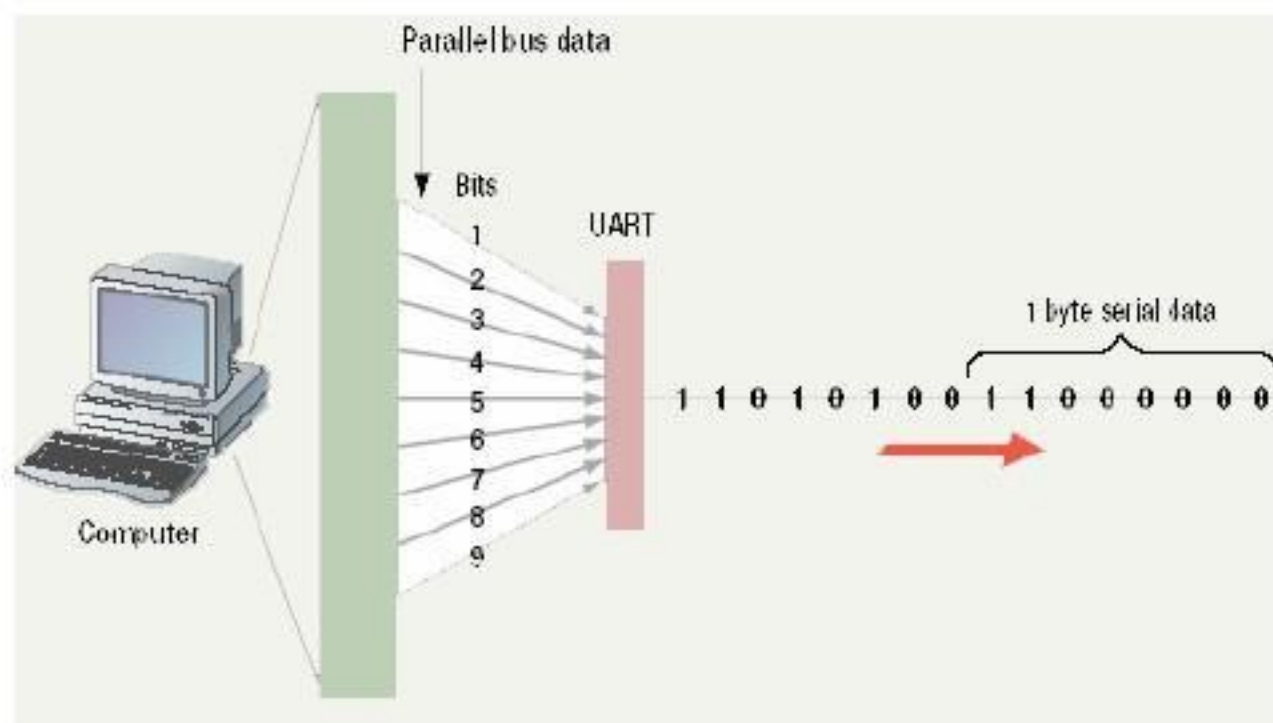
Universal Asynchronous Receiver/Transmitter

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Serial Communications



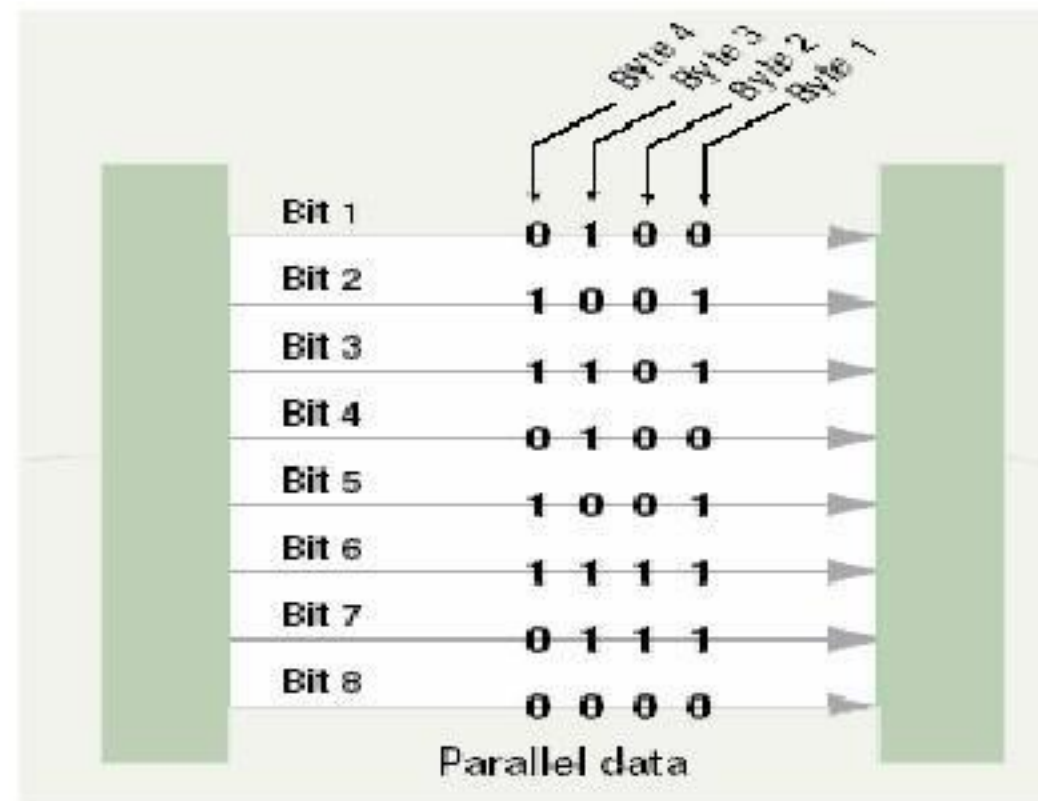
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Parallel Communications



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Functions of Operating Systems

- Provide a user interface
- Run programs
- Manage hardware devices
- Organized file storage

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Types of Operating Systems

- Real-time operating system
 - Very fast small OS
 - Built into a device
 - Respond quickly to user input
 - MP3 players, Medical devices

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Types of Operating Systems

- Single user/Single tasking OS
 - One user works on the system
 - Performs one task at a time
 - MS-DOS and Palm OS
 - Take up little space on disk
 - Run on inexpensive computers

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Types of Operating Systems

- Single user/Multitasking OS
 - User performs many tasks at once
 - Most common form of OS
 - Windows XP and OS X
 - Require expensive computers
 - Tend to be complex

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Types of Operating Systems

- Multi user/Multitasking OS
 - Many users connect to one computer
 - Each user has a unique session
 - UNIX, Linux, and VMS
 - Maintenance can be easy
 - Requires a powerful computer

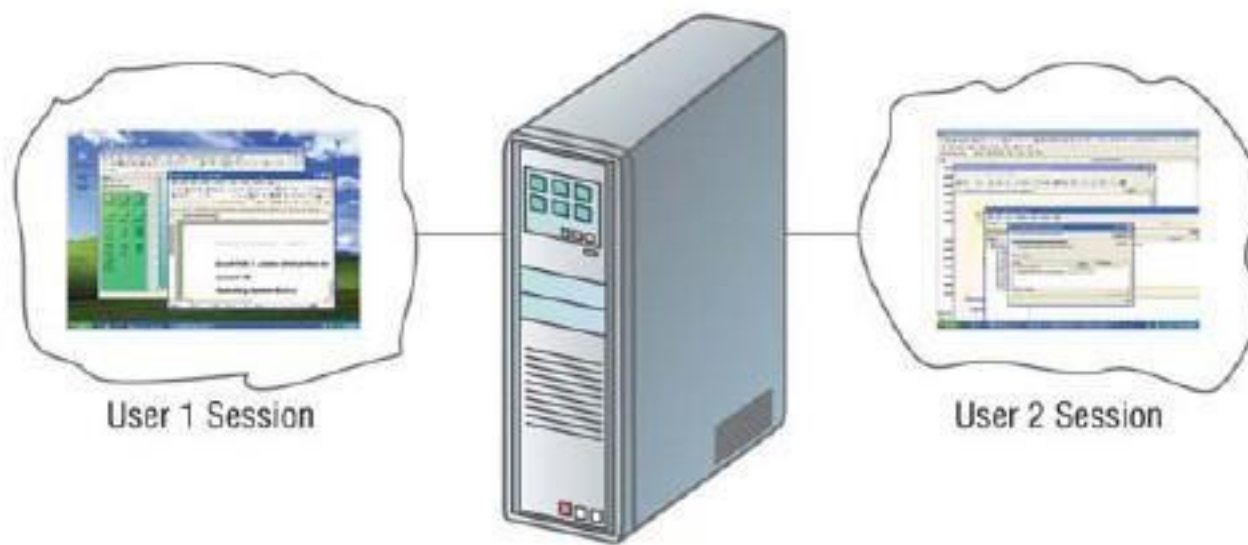
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Multi User / Multi Tasking OS



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Providing a User Interface

- User interface
 - How a user interacts with a computer
 - Require different skill sets

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Providing a User Interface

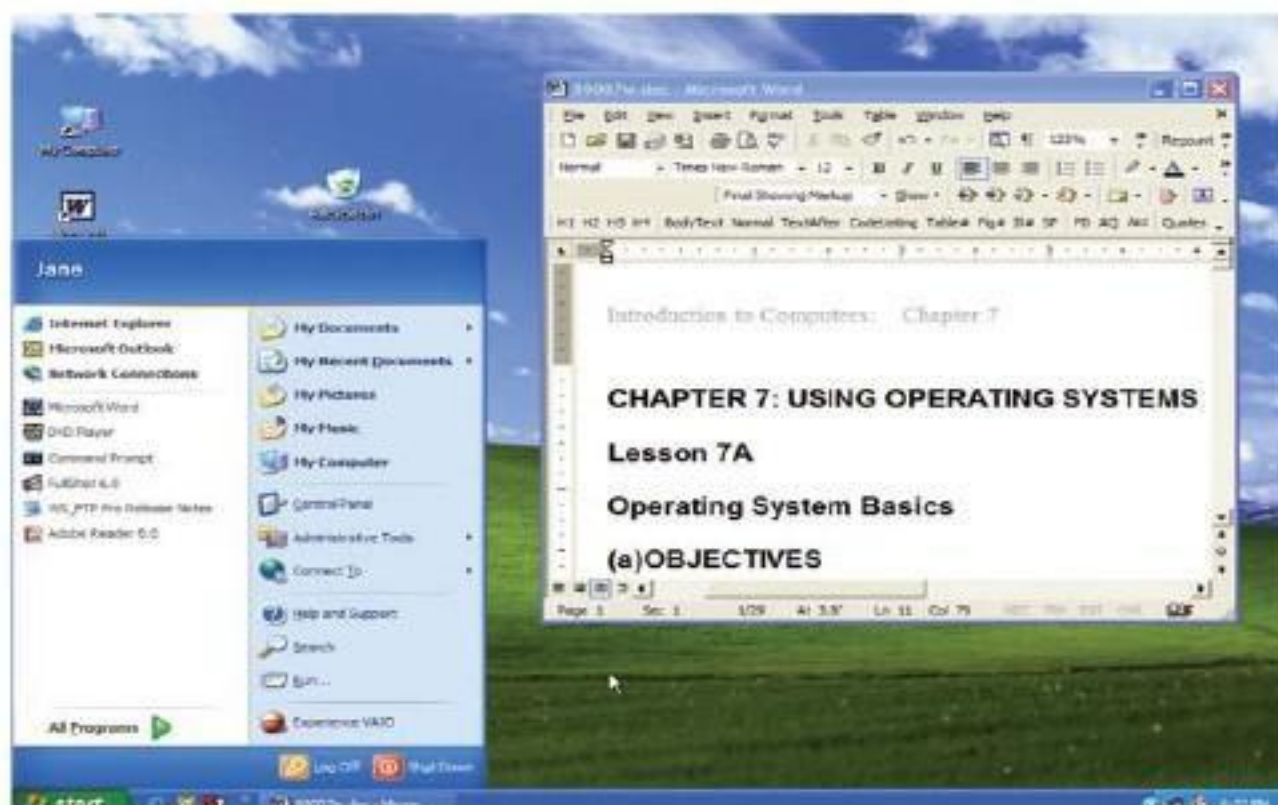
- Graphical user interface (GUI)
 - Most common interface
 - Windows, OS X, Gnome, KDE
 - Uses a mouse to control objects
 - Uses a desktop metaphor
 - Shortcuts open programs or documents
 - Open documents have additional objects
 - Task switching
 - Dialog boxes allow directed input

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Graphical User Interface



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Providing a User Interface

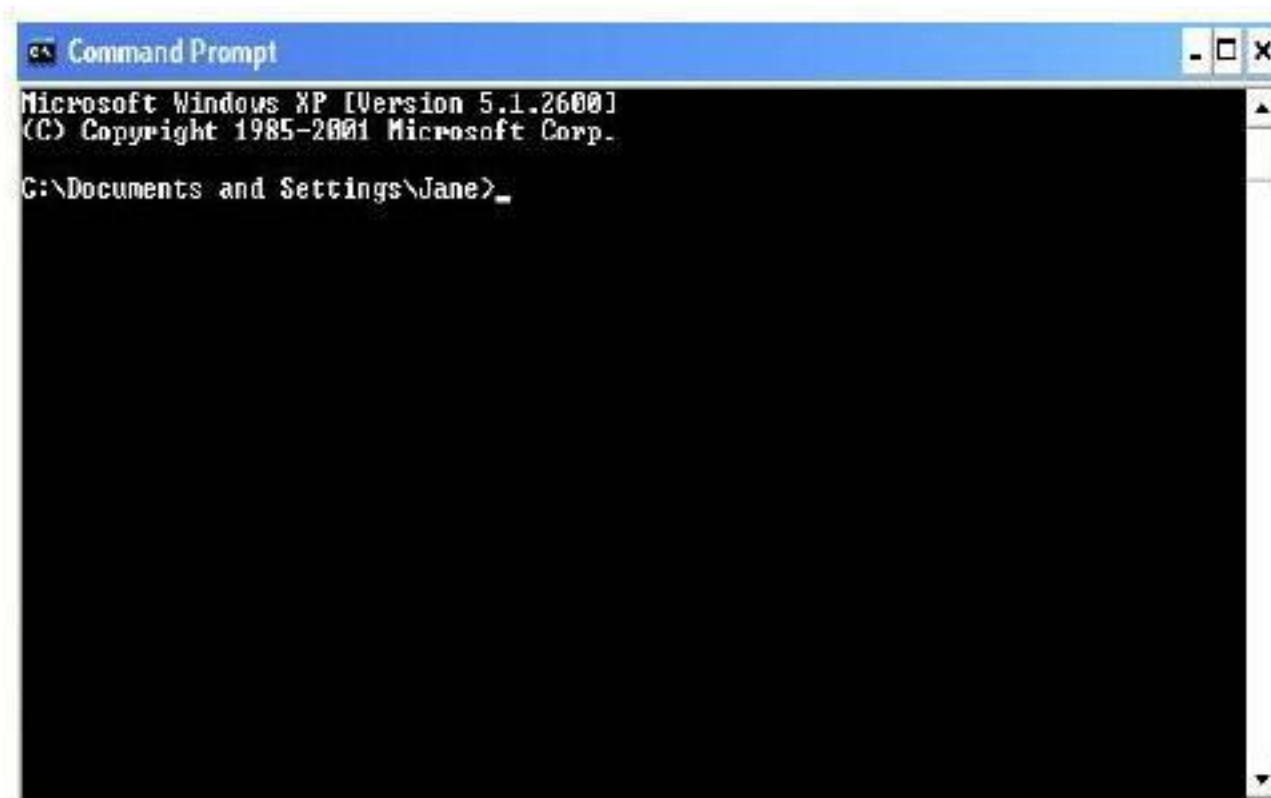
- Command line interfaces
 - Older interface
 - DOS, Linux, UNIX
 - User types commands at a prompt
 - User must remember all commands
 - Included in all GUIs

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Command Line Interface

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Organizing Files and Folders

- Organized storage
- Long file names
- Folders can be created and nested
- All storage devices work consistently

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PC Operating Systems

- Microsoft Windows is the most popular
 - Installed more than other OS combined
 - Installed on about 95% of computers
 - Apple and Linux represent the other 5%

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PC Operating Systems

- DOS (Disk Operating System)
 - Single user single-tasking OS
 - Command line interface
 - 16-bit OS
 - Powerful
 - Fast
 - Supports legacy applications

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PC Operating Systems

- Windows XP
 - Microsoft's newest desktop product
 - Different look from 2000
 - Many different versions
 - Digital multimedia support was enhanced
 - Communications was enhanced
 - Mobile computing became a priority

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PC Operating Systems

- Linux
 - Free or inexpensive version of UNIX
 - 32-bit OS
 - Very stable and fast
 - Most flavors are open source
 - X Windows GUI
 - Command line interface is available

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PC Operating Systems

- Macintosh operating systems
 - OS X
 - Based on FreeBSD Linux
 - Very stable and easy to configure
 - Only runs on Mac hardware

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Networking Operating Systems

- NT Server
 - Same core as NT Workstation
 - Security added
 - Multi user capability added
 - RAID support included

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Networking Operating Systems

- Windows 2003 Server
 - Designed to compete with UNIX
 - XP like interface
 - Better support for XP computers
 - MS .NET framework supported
 - Distributed programs are supported
 - Server Standard Edition
 - Data Center Edition

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Networking Operating Systems

- Linux for servers
 - Popular in small businesses
 - Stable, fast and inexpensive
 - Linux's popularity is growing

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Embedded Operating Systems

- Devices have EOS built in
- Cell phones, PDAs, medical equipment
- Stable and fast

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Embedded Operating Systems

- Windows CE
 - Not based on a desktop OS
 - Customized for each device
 - PDA and cell phones
 - Microsoft Automotive will run in cars

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Embedded Operating Systems

- Palm OS
 - Standard on Palm PDA
 - First PDA OS for consumers
 - Can be found on cell phones

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Embedded Operating Systems

View in AR



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Acquiring Software

- Commercial software
 - Software that must be purchased
 - Stand alone products
 - Solve one type of problem
 - Software suites
 - Integrated tools that work together
 - Solve many problems
- Shareware
 - Try before you buy
 - May deactivate if not purchased

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Acquiring Software

- Freeware
 - No obligation to purchase
 - Donations often accepted
 - Software may be distributed freely
 - Public domain software

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Acquiring Software

- Open source
 - Programs distributed with source code
 - Allows users to modify the software
 - Modifications and comments are welcome
 - Linux and OpenOffice

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Word Processing Programs

- Creates text documents
- Graphics and other objects are supported
- Professional quality can be achieved
- Simple web pages may be created

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Word Processing Programs

- Interface
 - Document area
 - Menu bar
 - Toolbars
 - Rulers
 - Scroll bars
 - Status bars

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Spreadsheet Programs

- Calculates numbers and finances
- Data viewable in many ways
 - Tables
 - Graphs
- Complex calculations can be automated

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Presentation Programs

- Creates slides or transparencies
- Complete set is a presentation
- Enhances a speech or lecture
- Color and animation enhance the slides

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Network Definition

- Set of technologies that connects computers
- Allows communication and collaboration between users

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The Uses of a Network

- Simultaneous access to data
 - Data files are shared
 - Access can be limited
 - Shared files stored on a server
 - Software can be shared
 - Site licenses
 - Network versions
 - Application servers

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The Uses of a Network

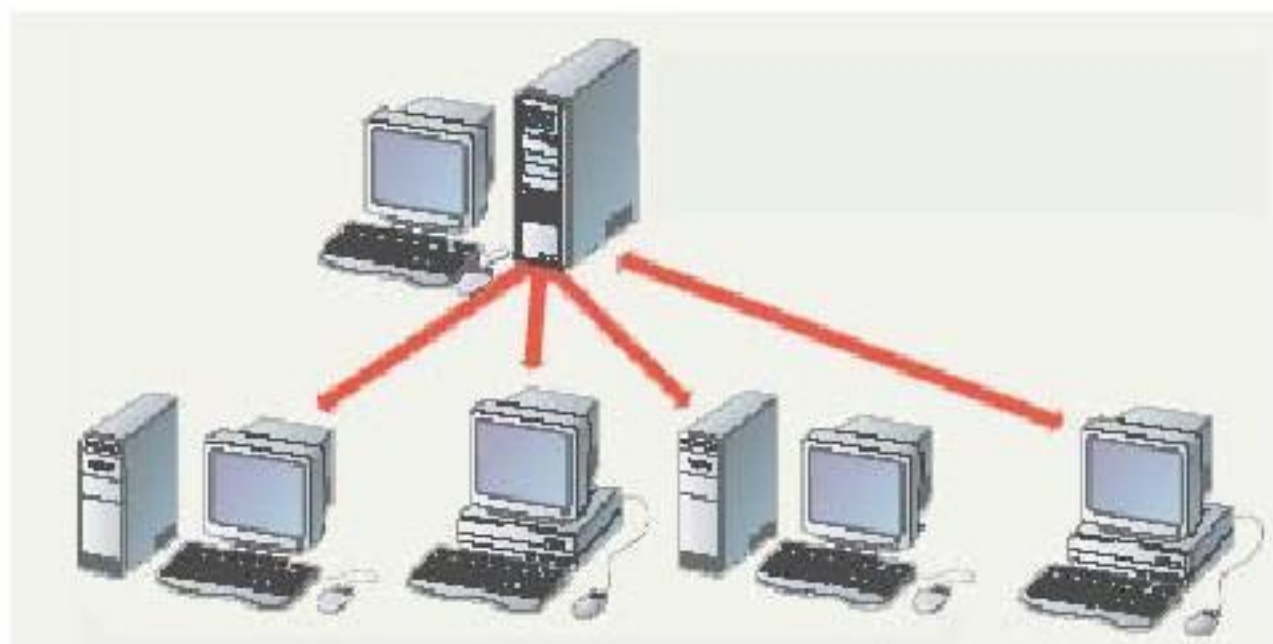
- Shared peripheral device
 - Printers and faxes are common shares
 - Reduces the cost per user
 - Devices can be connected to the network
 - Print servers control network printing
 - Manage the print queue

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Sharing Data



File server contains documents used by other computers.

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The Uses of a Network

- Personal communication
 - Email
 - Instantaneous communication
 - Conferencing
 - Tele conferencing
 - Videoconferencing
 - Audio-conferencing
 - Data-conferencing
 - Voice over IP
 - Phone communication over network wires

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The Uses of a Network

- Easier data backup
 - Backup copies data to removable media
 - Server data backed up in one step

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Common Network Types

- Local Area Network (LAN)
 - Contains printers, servers and computers
 - Systems are close to each other
 - Contained in one office or building
 - Organizations often have several LANS

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Common Network Types

- Wide Area Networks (WAN)
 - Two or more LANs connected
 - Over a large geographic area
 - Typically use public or leased lines
 - Phone lines
 - Satellite
 - The Internet is a WAN

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Hybrid Network Types

- Campus Area Networks (CAN)
 - A LAN in one large geographic area
 - Resources related to the same organization
 - Each department shares the LAN

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Hybrid Network Types

- Metropolitan Area Network (MAN)
 - Large network that connects different organizations
 - Shares regional resources
 - A network provider sells time

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Hybrid Network Types

- Home Area Network (HAN)
 - Small scale network
 - Connects computers and entertainment appliances
 - Found mainly in the home

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Hybrid Network Types

- Personal Area Network (PAN)
 - Very small scale network
 - Range is less than 2 meters
 - Cell phones, PDAs, MP3 players

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How Networks Are Structured

- Server based network
 - Node is any network device
 - Nodes are called clients
 - Nodes and servers share data roles
 - Servers control what the node accesses
 - Users gain access by logging in
 - Server is the most important computer

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Network Media

- Links that connect nodes
- Choice impacts
 - Speed
 - Security
 - Size

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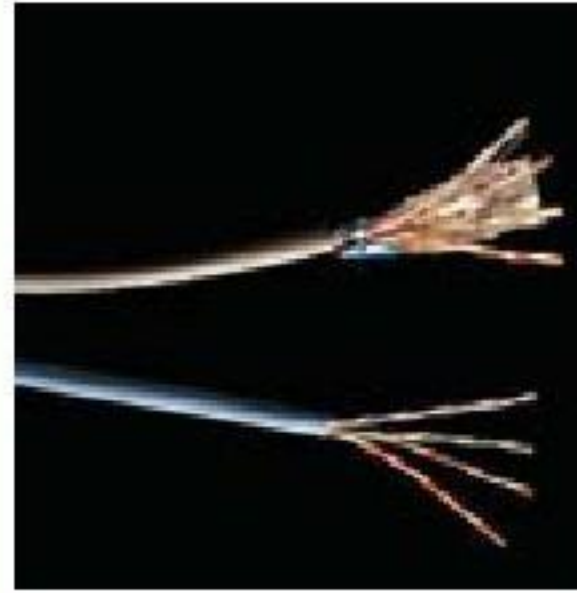
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Wire Based Media

- Twisted-pair cabling
 - Most common LAN cable
 - Called Cat5 or 100BaseT
 - Four pairs of copper cable twisted
 - May be shielded from interference
 - Speeds range from 1 Mbps to 1,000 Mbps



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Wire Based Media

- Coaxial cable
 - Similar to cable TV wire
 - One wire runs through cable
 - Shielded from interference
 - Speeds up to 10 Mbps
 - Nearly obsolete

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Wire Based Media

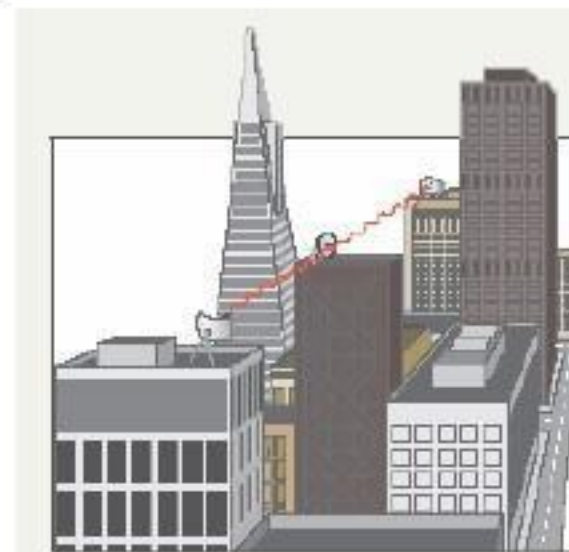
- Fiber-optic cable
 - Data is transmitted with light pulses
 - Glass strand instead of cable
 - Immune to interference
 - Very secure
 - Hard to work with
 - Speeds up to 100 Gbps



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Wireless Media

- Data transmitted through the air
- LANs use radio waves
- WANs use microwave signals
- Easy to setup
- Difficult to secure



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Network Hardware

- Network interface cards
 - Network adapter
 - Connects node to the media
 - Unique Machine Access Code (MAC)



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Network Hardware

NIC01
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Network Hardware

- Hubs
 - Center of a star network
 - All nodes receive transmitted packets
 - Slow and insecure

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Network Hardware

- Switches
 - Replacement for hubs
 - Only intended node receives transmission
 - Fast and secure

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Network Hardware

- Bridge
 - Connects two or more LANs together
 - Packets sent to remote LAN cross
 - Other packets do not cross
 - Segments the network on MAC addresses

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Network Hardware

- Router
 - Connects two or more LANs together
 - Packets sent to remote LAN cross
 - Network is segmented by IP address
 - Connect internal networks to the Internet
 - Need configured before installation

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Network Hardware

Router



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Network Protocols

- Language of the network
 - Rules of communication
 - Error resolution
 - Defines collision and collision recovery
 - Size of packet
 - Naming rules for computers

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Network Protocols

- TCP/IP
 - Transmission Control Protocol/Internet Protocol
 - Most popular protocol
 - Machines assigned a name of 4 numbers
 - IP address
 - 209.8.166.179 is the White House's web site
 - Dynamic Host Configuration Protocol
 - Simplifies assignment of IP addresses
 - Required for Internet access

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Network Protocols

- IPX/SPX
 - Internet Packet Exchange/Sequenced Packet Exchange
 - Older protocol
 - Associated with Novell Netware
 - Replaced by TCP/IP

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Modem Communications

- Plain Old Telephone System (POTS)
 - Standard phone line
 - Two-way voice communication
 - Uses analog transmission techniques
 - Data communication is slow

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Modem Communications

- Modems
 - Modulator/Demodulator
 - Modulator converts digital to analog
 - Speed measured in bits per second (bps)
 - Current fastest speed is 56 Kbps
 - Quality of phone lines dictates speed
 - V.92 is the current modem standard
 - Several modem types
 - Internal , External , Voice ,Fax

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Modem Communications

- Modem uses
 - Connection to the Internet
 - File transfer
 - Uploading
 - Downloading



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Modem Communications



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Digital Data Connections

- Broadband connection
 - Any data connection faster than 56 Kbps
 - Common in business
 - Becoming popular in home installations

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Digital Data Connections

- ISDN lines
 - Integrated Services Digital Network
 - Basic rate uses three channels
 - Two data channels each support 64 Kbps
 - Error correction channel
 - Primary rate uses 24 data channels

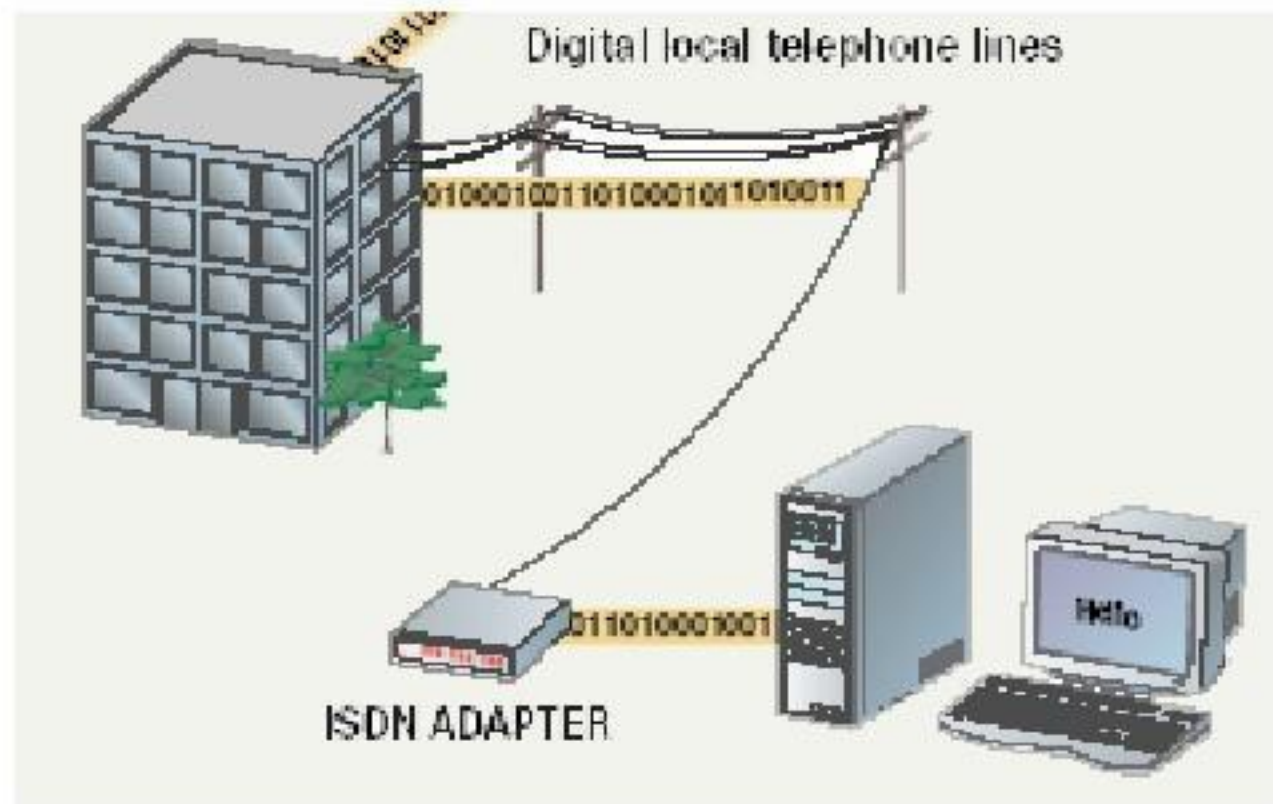
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ISDN Connection



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ISDN Connection



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Digital Data Connections

- DSL technologies
 - Digital Subscriber Line
 - Popular with home users
 - Speeds range from 100 Kbps to 30 Mbps
 - Asymmetrical DSL (ADSL)
 - Upload speed slower than download speed
 - Symmetrical DSL (SDSL)
 - Requires a DSL modem

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Digital Data Connections

- Cable modem connections
 - Popular with home and office users
 - Speeds between 1 and 3 Mbps
 - Requires a cable modem

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Wireless Networks

- Benefits
 - No cable to pull
 - Mobile devices access network resources
 - Mobility and flexibility for office workers

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Wireless Networks

- Wireless 802.11
 - Also called Wi-Fi
 - IEEE standard
 - Institute of Electronic and Electrical Engineers
 - Several versions
 - 802.11b connects up to 11Mbps
 - 802.11g connects up to 56Mbps
 - Use the same type of devices

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Wireless Networks

- Wireless Access Point
 - Center of a wireless network
 - WAPS combined cover a larger area
 - Distance to WAP determines bandwidth
 - Range is 50 to 150 meters
 - Extension points can extend range



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Wireless Networks

Router



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Wireless Networks

- Wireless Adapters
 - Used by devices to connect
 - Includes signal strength software



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Wireless Networks

Wireless
Adapter
CF



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Wired Internet Connections

- Dial-up connections
 - Standard phone lines and a modem
 - Computer dials a number for your ISP
 - All Internet programs use the connection
 - Applications might need configured

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Wired Internet Connections

- High-speed broadband connections
 - Any connection faster than dialup
 - Networks share the broadband connection
 - All users access the same connection
 - T or DSL lines common
 - Home use increasing due to
 - Lower cost
 - Increased availability

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Wired Internet Connections

- ISDN
 - Integrated Services Digital Network
 - Offers speeds up to 1.5 Mbps
 - Uses standard phone lines
 - Requires special equipment
 - Simultaneous use of phone and data

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Wired Internet Connections

- DSL
 - Offers speeds up to 30 Mbps
 - Uses modified phone lines
 - Needs special DSL modem
 - Simultaneous use of phone and data
 - Asynchronous DSL
 - Different up and download speeds
 - Synchronous DSL
 - Variable DSL
 - Speed changes based on traffic

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Wired Internet Connections

- Cable modems
 - Speeds up to 3 Mbps
 - Uses cable TV wires
 - Requires a cable modem
 - Simultaneous use of TV and data

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Software Accessing The Internet

- Application programming interface (API)
 - Simplifies connection to network devices
 - Allows any application to access Internet
 - UNIX API uses sockets
 - Windows API uses winsock
- Network drivers
 - Control access to the hardware
 - Ethernet and dial-up require a driver
 - API connects to the driver

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Wireless Internet Connection

- Satellite services
 - Internet access in remote regions
 - Suitable for home and office use
 - Needs a VSAT at the client
 - Very Small Aperture Terminal
 - Connects to the satellite
 - Modem connects to the VSAT

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Wireless Internet Connection

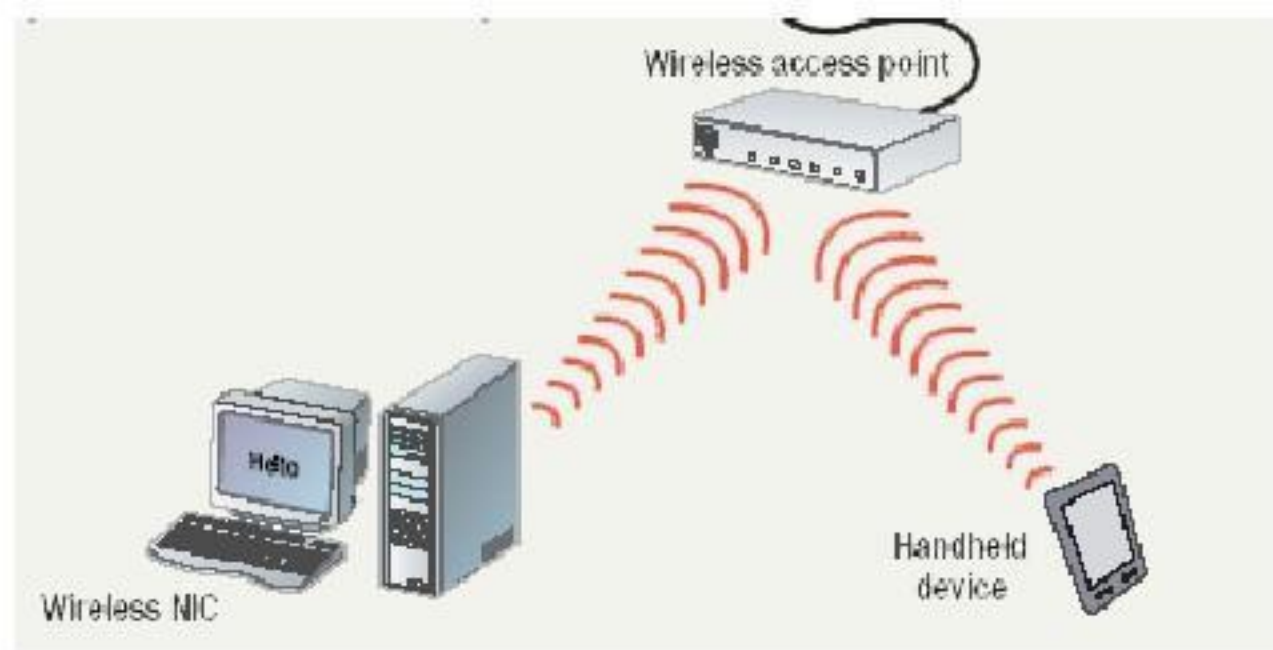
- Wireless LANs (WLAN)
 - Network without wires
 - Connects to a broadband LAN connection
 - WAP are wired directly into the LAN
 - Nodes use wireless

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Wireless LAN

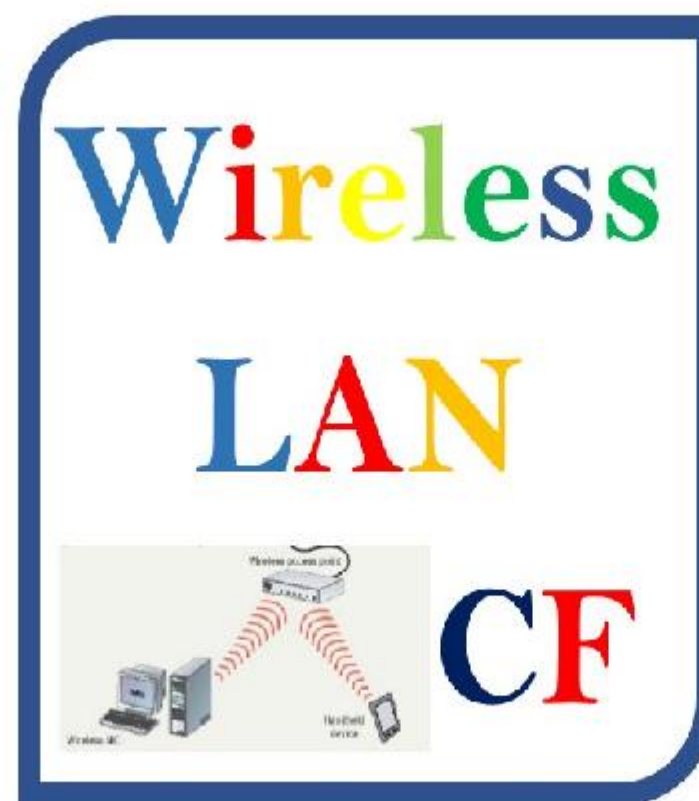


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Wireless LAN



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The Internet

- Internet is an international network connection of over 400,000 networks in more than 150 countries
- While English dominates current sites, Chinese is predicted to overtake English in sites by 2015
- Many sites have translated attached sites or provide simultaneous translation

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Intranet & Extranet

- *Intranets*
 - network within the organization
 - security behind firewalls
- *Extranet*
 - extended intranets outside to employees such as salespeople
 - extended to suppliers, B-2-B customers, and alliances

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Internet Tools

- World Wide Web
- E-mail
- Usenet newsgroups
- Mailing lists
- FTP, file transfer protocol
- Telnet, allow logon as a client computer

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Web Address

- *URL*
 - Uniform Resource Locator
- *http://*
 - hypertext transfer protocol
- *ftp://*
 - file transfer protocol

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Internet Formats

- *HTML*
 - Hypertext Markup Language
 - html
 - Microsoft variants
 - htm
 - asp
 - active server pages
- *XML*
 - Extensible Markup Language

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Internet Address

- *Domain Name System*
 - Domain
 - user ID, subdomain, domain (location), root domain (domain type), and country
 - com, edu, gov, mil, net, org
- Visit the Internet Assigned Numbers Authority
 - <http://www.iana.com>
 - \$75 for a limited time

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Consumer E-Commerce

- Online shopping
 - Purchasing through a web site
 - Also called Business to Consumer (B2C)
 - List of products is nearly endless
 - Features
 - Searchable catalog
 - Secure checkout
 - Customer support page

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Consumer E-Commerce

- Online banking
 - Most teller services handled online
 - Balances can be obtained
 - Bills can be paid
 - Access is typically very secure
 - Not available at all banks

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Business E-Commerce

- Business to business (B2B)
 - Business transaction between companies
 - Accounts for the majority of e-commerce
 - Initially private networks were used
 - Common transactions
 - Reorder inventory
 - Purchase stocks
 - Credit checks

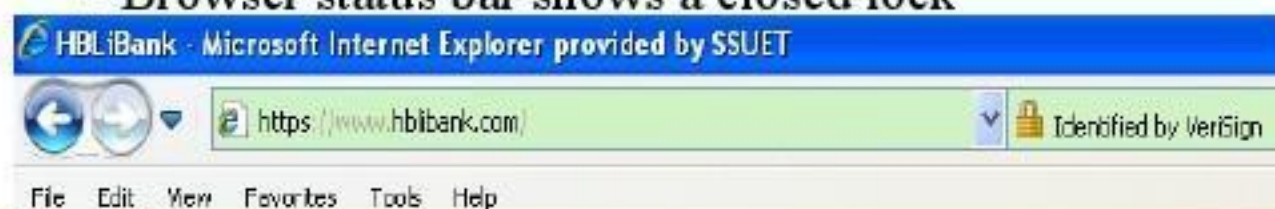
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Security

- Secure web pages
 - Allows safe transmission of credit card
 - Use the secure socket layer (SSL)
 - Encrypts data transmission to the server
 - Secure pages are easily identified
 - URL uses https://
 - Browser status bar shows a closed lock



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